Backtesting Report

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1. Strategy Overview

The strategy used in this report uses the Williams %R indicator and builds thresholds around it. Equation 1 shows the equation for calculating Williams %R. The rules for trading are as follows[2]:

1. We enter a long position when the indicator crosses below -90
2. Exit the long position when the indicator crosses above -30 or when the close is higher than yesterday's high

$$\%\mathrm{R} = -100((\text{last n days high–close})/(\text{last n days high–last n days low})) \tag{1}$$

There are a few tunable parameters for this strategy. There are the two levels that the indicator crossing will cause a position change. For this strategy we used the two levels shown above. The third parameter is the lookback period for determining the previous high and low.

2. Strategy History

The William's %R indicator was originally developed by trader named Larry Williams in 1973[1, 3]. The goal of the indicator is to measure and quantify overbought and oversold securities. Readings between 0 and -20 and typically referred to as overbought and reading between -80 and -100 are typically referred to as oversold. The originally formula printed in William's book actually multiplied the formula by a positive 100, however, it has since been multiplied by -100 and proliferated as such.

In his publication, William utilized a lookback period of 10 trading days and did not directly trade on the oversold and overbought thresholds. He instead utilized the following rules:

1. Buy On:
   (a) %R reaches -100%
   (b) Five trading days pass since -100% was last reached
   (c) %R rises above -95% or -85%.
2. Sell On:
   (a) %R reaches 0%
   (b) Five trading days pass since 0% was last reached
   (c) %R drops below -5% or -15%

3. Backtesting

All data was collected from Yahoo! Finance as daily data. I collected both S&P500 and USDEUR data for their entire lifetimes. I utilized the Backtrader library to implement my backtesting and changed their Williams %R formula to match the formula described for the original strategy.

3.1 Original Strategy Performance

Figure 1 shows the equity curve of the original strategy performance on the SPY. They tested over the entire lifetime of the SPY. This strategy used the same rules I described above and they tested lookback periods of 2 to 25. This curve shows their best performance which was using a lookback period of 2. Their worst performance was using a lookback period of 25 which leads me to believe that the strategy gets progressively worse with longer lookback periods.

The statistics for their backtest are as follows:

- Market exposure: 22%
- Number of trades: 598
- Average gain per trade: 0.6%
- Max drawdown: 17%
- Profit factor: 2.2
- Risk adjusted return: 52%

They also noted that it performed exceptionally well during the GFC in 2008/09 and during Covid. In 2008 and 2020 it had a 98.9% and 43.2% returns respectively as well as a 15.7% return during the bear market of 2022.

3.2 Reproduced Strategy Performance

Figure 2 shows the equity curves of my implementation of Williams %R on SPY as well as the buy and hold performance on SPY. This strategy used the same 2-day lookback as the original implementation. Similarly to the original backtest, this strategy also invested all of its cash into every position. I also used a 0% commission on each trade to match the original implementation and starting cash of $10,000.

The statistics for my backtest are as follows:

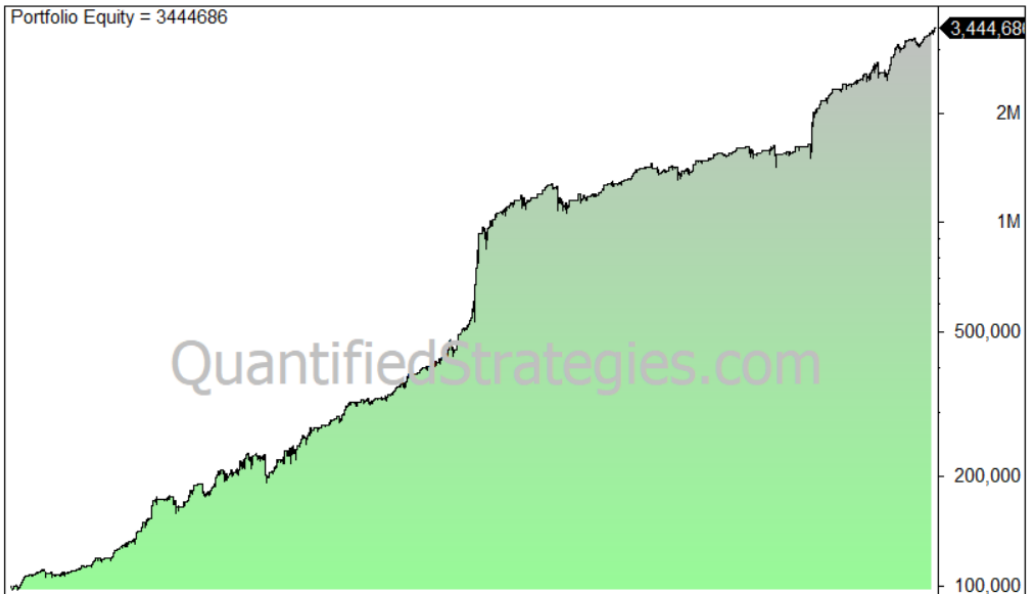


Figure 1: Equity curve of original strategy performance on the S&P500 using a lookback of 2 days.

- Market exposure: 11.22%
- Number of trades: 272
- Average gain per trade: 0.62%
- Max drawdown: 17.49%
- Profit factor: 2.1
- Risk adjusted return: 47.62%

These numbers are very similar to the original strategy. There are much fewer trades in my implementation than in the original implementation. This could be a result of how they actually perform the trades. In some cases, my backtest will receive a signal to buy but the order to buy won't execute due to the open of the next day being very different than the previous close. It is not made clear how the original strategy make their trades, but if they are trading on a close price this could lead to many more trades.

Other than number of trades, the rest of the statistics are very similar. This does result in a much lower CAGR of 5.34% compared to the 11.9% CAGR they reported. They also reported a CAGR of 10.3% for the buy and hold strategy on SPY while I only calculated a 8.51% CAGR. This is likely

a result of them reinvesting dividends which is not something I included in my buy and hold strategy.

This performance provides good evidence that this is a viable strategy due to the extremly low market exposure. However, to actually make that conclusion I would want to run more tests with various lookback periods and position thresholds. This would allow me to know if the strategy has real predictive capability or if I just stumbled upon a combination that worked. With that being said, this initial backtest performs very well and beat my expectations. While it does not beat the buy and hold of SPY, it is only in the market 11% of the time allows you to make other trades in the remaining 89% where your money is not tied up. If you buy and hold your money is tied up 100% of the time.

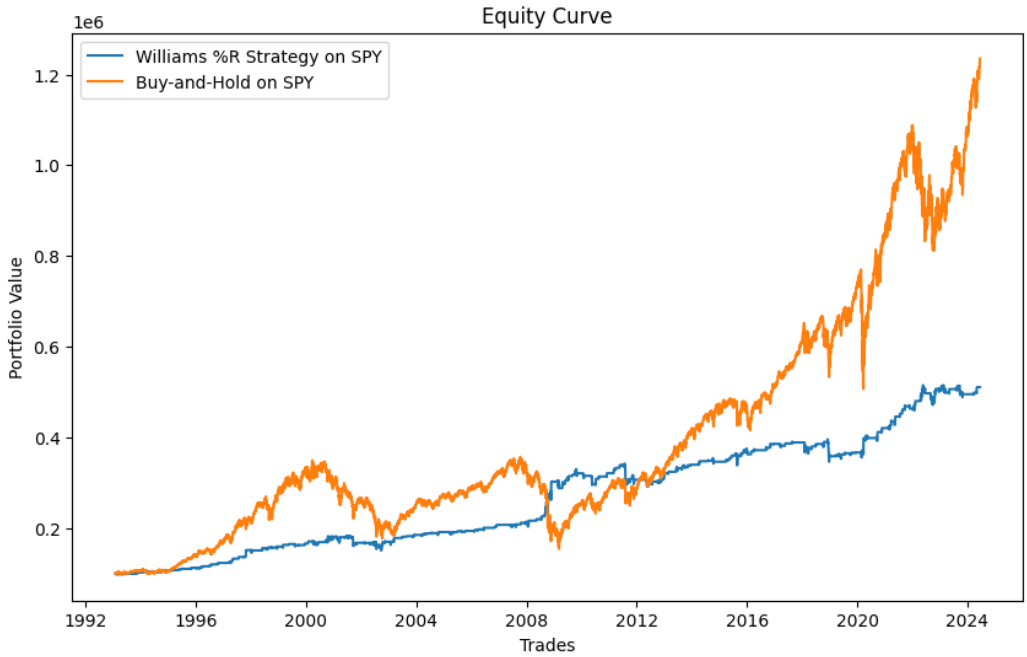


Figure 2: Equity curve of my implementation performance on SPY using a lookback of 2 days as well as the buy and hold of SPY.

3.3 Optimization Universe

After evaluating the performance of this strategy using the best parameters found by the original poster, I built an optimization universe to see if I can find even better parameters. I introduce three parameters to search through: lookback period, upperband threshold, and lowerband threshold. I evaluated

each parameter in the following ranges:

- Lookback period: 1 to 20 in increments of 1
- Upperband threshold: -10 to -35 in increments of -5
- Lowerband threshold: -70 to -95 in increments of -5

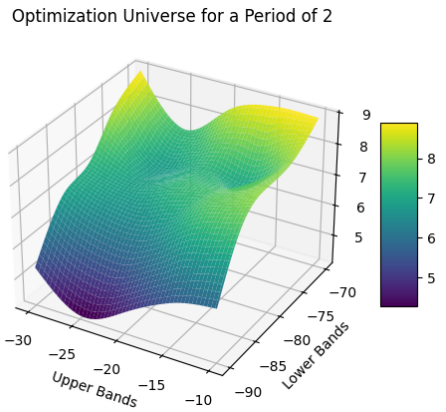
Figure 3 shows the annualized returns of each set of upper and lower bands for selected lookback periods 2, 5, 10, and 20. The data is interpolated using cubic splines to get the graphs shown, however, the averages are generated only from the CAGR from the strategy. All sets of parameters show a positive annualized return in the range of about 3.5 to about 8.5. As shown, a lookback period of 2 has highest average annualized returns of 7.06% which got progressively worse as the lookback period increased down to a lookback period of 20 as shown in Figure 4. This matches the original strategy analysis which also found a lookback period of 2 to be optimal and found the worst performance with the maximum lookback distance tested.

The original strategy did not evaluate performance over various upper and lower bands. As shown in Figure 3, the best performance appears at the higher lower band values and the worst performance occurs at the lower lower bands values. Additionally, outside of Figure 3a, changing the upper band value does not affect the overall CAGR by much. Even in Figure 3a changing the value of the lower band has a much larger affect on the CAGR. In each of the graphs, they only flat area where we can have some confidence in the future performance would be a lower band value between -70 and -75.

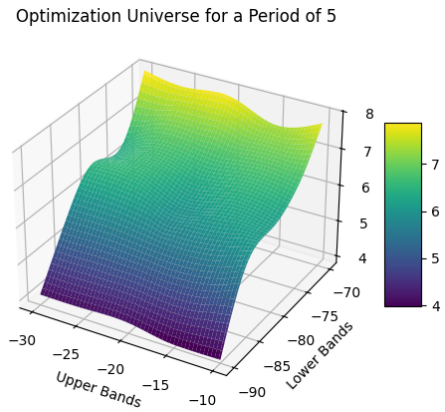
3.4 Optimal Parameters

Figure 5 shows the equity curve again with the new optimal parameters found from the optimization universe. It achieves a CAGR of 7.91%. The following is the performance of this strategy using the new paramerters:

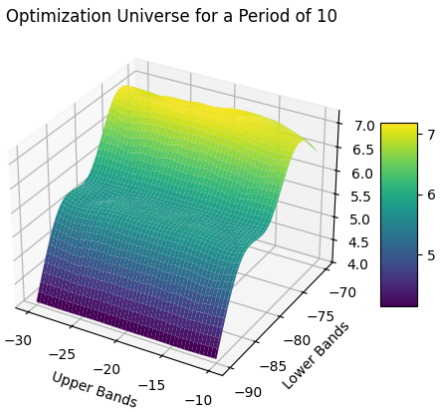
- Market exposure: 23.62%
- Number of trades: 516
- Average gain per trade: 0.49%
- Max drawdown: 22.89%
- Profit factor: 1.8
- Risk adjusted return: 33.49%



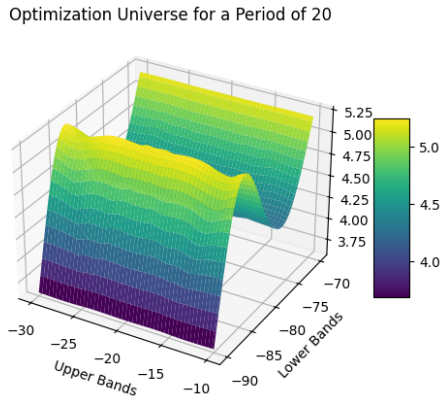
(a) Period of 2, Avg CAGR: 7.11%



(b) Period of 5, Avg CAGR: 6.04%



(c) Period of 10, Avg CAGR: 5.87%



(d) Period of 20, Avg CAGR: 4.58%

Figure 3: Optimization universe for selected values of lookback period.

While the overall performance profit increased with the new parameters, if we dive deeper into the statistics of the strategy is does not look as good. The market exposure is over two times as high, the max drowdown is higher, and the average gain per trade is lower. This leads me to believe that there are more metrics than just profit that need to be analyzed when looking at an optimization universe. These new parameters do outpace the buy and hold strategy for a while but in the end still fall slightly short of the buy and hold performance of 8.52% CAGR.

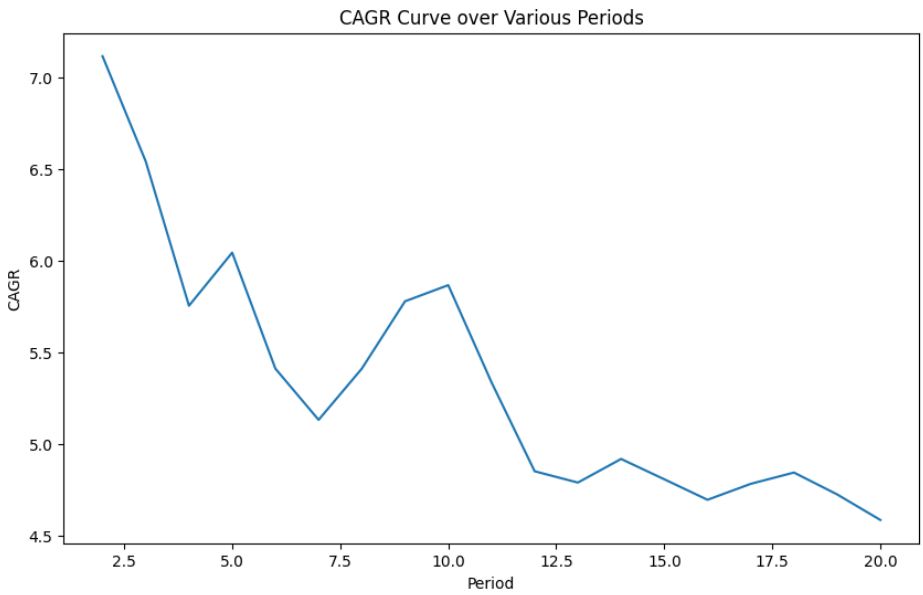


Figure 4: CAGR over various periods.

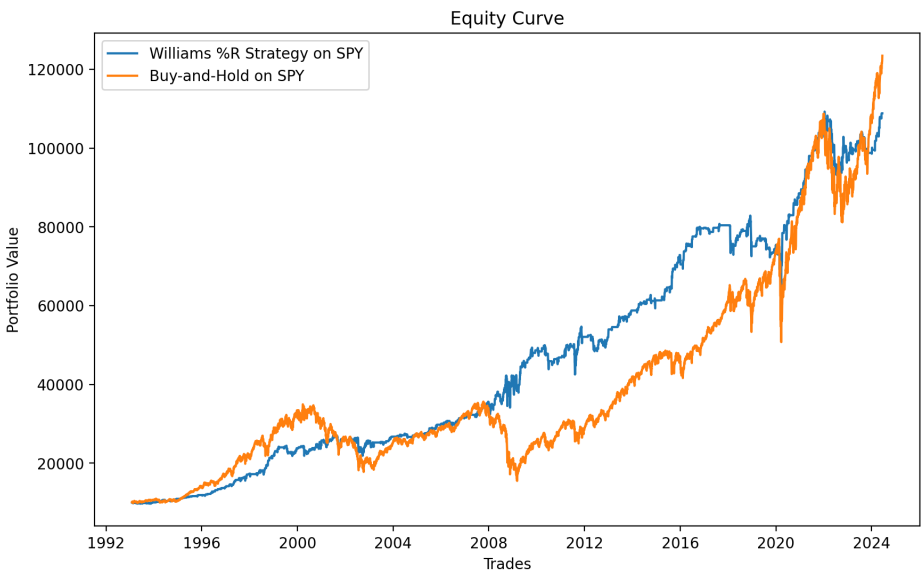


Figure 5: Equity curve of my implementation performance on SPY using a lookback of 2 days, lowerband of -72.5, and an upperband of -20 as well as the buy and hold of SPY.

4. References

[1] Wikimedia Foundation. Williams %r. https://en.wikipedia.org/wiki/Williams_%25R#:~:text=It%20was%20developed%20by%20a,Williams%2C%20is%20a%20momentum%20oscillator., March 16 2024. Wikipedia.

[2] O. Groette. Williams %r trading strategy - 81% win rate - williams percent range vs rsi (backtest, indicator & systems). https://www.quantifiedstrategies.com/williams-r-trading-strategy/, June 15 2024. Quantified Strategies.

[3] L. Williams. Original williams %r. williams percent r indicator %r, n.d.